THE TRANSPORT CONJECTURE BETWEEN LATTICE SU(N) GLUEBALLS AND BIANCHI ORBIFOLD SPECTRA: A SIX-STAGE STRUCTURAL MAP AND A WILES-STYLE PROVED-CONDITIONAL CLOSURE ON TWO NAMED AXIOMS (V3)

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Abstract. The Transport Conjecture asserts the existence of a map Φ from the continuum limit of the lattice SU(N) Wightman scalar glueball mass $m_{0^{++}}^{\mathrm{lat}}(\mathrm{SU}(N))/\sqrt{\sigma_N}$ to a spectral observable $\sqrt{\lambda_1(\Delta_{\chi_*})}$ on the Bianchi 3-orbifold Y_K with $K = Y_K^{\mathrm{CR}}(N)$ the Center–Rank canonical representative [Ŕ26a], such that the dimensionless ratio equals the universal product $\sqrt{2\pi e}\cdot\sqrt{2/3}\cdot F(N)$ of [Ŕ26d]. We replace the four-strategy open-problem statement of V1–V2 with a single *structural* factorisation of Φ as a six-stage composition between categorically-typed objects, and we isolate *exactly two* named open axioms required for closure: (T_1), the Osterwalder–Schrader reconstruction for the 4D lattice continuum limit of SU(N) Yang–Mills (equivalent to the Clay Yang–Mills Millennium problem [JW00] for the surrogate observable); and (T_2), the spectral identification Φ_4 from the 4D Wightman 2-point function to the Cl(K)-equivariant heat trace on Y_K. Modulo (T_1) and (T_2), the remaining four stages $\Phi_1,\Phi_2,\Phi_3,\Phi_5,\Phi_6$ are unconditional and proved in the cited companion papers. Our Theorem 1.3 is therefore a Wiles-1995-style *PROVED-CONDITIONAL* closure on (T_1) and (T_2) : a clean, single conditional statement of the Transport Conjecture. Lattice empirical agreement at RMS $\sim 1.3\%$ across the nine canonical Athenodorou–Teper 2021 anchors $N \in \{2,3,4,5,6,8,10,12,\infty\}$ (Tab. 3) is consistent with, but does not prove, the conjecture; the chi-squared per anchor is $\chi^2/\mathrm{dof} \approx 1.70$ and the Bayes-factor comparison against a constant-gap null exceeds 10^{130} (§5). We retain the V2 Boolean falsifier ($\lambda_1 = 1$ on the χ_*-isotypic component) as the primary empirical test and propose it at $N = 3$, $K = \mathbb{Q}(\sqrt{-67})$, as the highest-ROI next experiment. A new section (§6) records five additional empirical tests supporting an entropy-saturation reading of the universal product, and Kevin's 3-layer physical derivation (§7) is added with explicit layer-by-layer honest caveats.

1. Introduction and main theorem

1.1. Statement of the problem. Let $N \geq 2$. The Athenodorou–Teper 2021 lattice continuum data [AT21, Tab. 34] give the dimensionless ratio $m_{0^{++}}^{\mathrm{lat}}(\mathrm{SU}(N))/\sqrt{\sigma_N}$ to within $\sim$ 1–3% for the eight finite ranks $N \in \{2,3,4,5,6,8,10,12\}$ and the $N \to \infty$ limit (the AT2021 paper does not cover $N = 7,9,11$). Over the past sequence of crossed-cosmos preprints [Ŕ26e, Ŕ26c, Ŕ26d, Ŕ26a, Ŕ26b, Ŕ26g], this ratio has been shown to equal, modulo a single conjectural identification (the *Transport Conjecture*), the universal product

$$\mathcal{C}(N) \;:=\; \sqrt{2\pi e}\cdot\sqrt{\tfrac{2}{3}}\cdot F(N), \qquad F(N) = \tfrac{9}{10}\,\frac{N^2+1}{N^2}\,. \tag{1}$$

Date: May 20, 2026.

2020 *Mathematics Subject Classification.* Primary 81T13, 81T25; Secondary 11F72, 58J50, 81T08.

Key words and phrases. Yang–Mills, lattice gauge theory, Bianchi 3-orbifold, Wightman, constructive QFT, Osterwalder–Schrader reconstruction, Selberg pretrace, Transport Conjecture, Wiles-style conditional, two-Millennium-stack.

ORCID 0009-0008-2443-7166. This work is part of the *crossed-cosmos* project (concept DOI 10.5281/zenodo.19686398). **V3 deltas vs V2** [Ŕ26g]**:** (i) the transport map Φ is now structurally factored as a six-stage composition $\Phi = \Phi_6 \circ \cdots \circ \Phi_1$ between named mathematical categories, with each stage either UNCOND from an existing companion theorem or attributed to one of two precisely identified open axioms; (ii) Theorem 1.3 is stated as a single *PROVED-CONDITIONAL* closure, Wiles-1995-style, on the named open axioms (T_1) (*OS-4D reconstruction for lattice SU(N)*) and (T_2) (*spectral identification Φ_4 from the 4D Wightman 2-point function to the* Cl(K)*-equivariant Bianchi heat trace*); (iii) the rest of the chain (stages $\Phi_1,\Phi_2,\Phi_3,\Phi_5,\Phi_6$) is unconditional and proved in the cited companion papers.

The RMS deviation over the nine AT2021 anchors is $\sim 1.3\%$ (Tab. 3). The factors $\sqrt{2\pi e}$ (Karamata–Stirling on $\mathbb{H}^3$), $\sqrt{2/3} = \sqrt{\xi^*(K)}$ (universal topological invariant of every Bianchi 3-orbifold [R26e]), and $F(N)$ (Dijkgraaf–Witten genus expansion factor at leading order $1+O(1/N^2)$) are individually understood : the first as a TIER-3 sketch [R26d, §4], the second as TIER 1 UNCOND [R26e], and the third as TIER 1 UNCOND for the *structural form* $c \cdot (1 + 1/N^2)$ from the classical genus expansion [tH74]. The *specific coefficient* $c = 9/10$ is an SU(3)-anchor calibration (Remark 2.2 below).

What is open — the Transport Conjecture proper — is the existence of a structural map Φ between the lattice SU(N) Wightman observable on $\mathbb{T}^4$ and a spectral observable on the Bianchi 3-orbifold Y_K with $K = Y_K^{\mathrm{CR}}(N)$. We articulate this map in §2 as a six-stage composition between named categorically-typed objects, identify two open axioms required for closure, and state the conjecture as a Wiles-1995-style *PROVED-CONDITIONAL* theorem on those axioms.

1.2. **The six-stage transport map.** The transport map Φ is the composition of six maps $\Phi_1, \dots, \Phi_6$ between the following typed objects (commutative diagram (3) below):

- $\mathcal{L}(N,a)$: the lattice SU(N) Wilson ensemble on $\mathbb{T}^4$ at lattice spacing a (well-defined for every $a > 0$);
- $\mathcal{W}^E_{a\to 0}(N)$: the continuum limit of the lattice Wightman 2-point function of the lowest scalar glueball operator $\mathcal{O}^{(0^{++})}$, considered as a *Euclidean* Schwinger function on $\mathbb{T}^4$;
- $\mathcal{W}^M(N)$: the Minkowski Wightman 2-point function on $\mathbb{T}^4$, obtained from $\mathcal{W}^E$ via Osterwalder–Schrader reconstruction [OS73, OS75];
- $\mathcal{H}_{0^{++}}(N)$: the closed gauge-invariant Hilbert space generated by the lowest scalar glueball, with self-adjoint Hamiltonian H_N and ground-state vector Ω_N;
- $\mathcal{T}_K^{(\chi_*)}$: the χ_*-isotypic component of the heat trace $\mathrm{Tr}_{L^2_{\mathrm{cusp}}(Y_K)}\, e^{-t\Delta}$ on the Bianchi 3-orbifold Y_K with $K = Y_K^{\mathrm{CR}}(N)$, as per the per-character decomposition of [R26c];
- $\xi^*(K)\sqrt{\lambda_1(\Delta_{\chi_*})}F(N)$: the right-hand side of (1), written suggestively as a triple product of the (UNCOND) universal constants $\sqrt{2\pi e} \cdot \sqrt{\xi^*(K)}$ and the topological combinatorial factor $F(N)$.

The maps Φ_i are defined in §2; we anticipate the table here as Table 1.

Stage	Domain $\to$ codomain	Description	Tier
Φ_1	$\mathcal{L}(N,a) \to \mathcal{W}^E_{a\to 0}(N)$	continuum limit at fixed physical volume	TIER 1 (folklore)
Φ_2	$\mathcal{W}^E_{a\to 0}(N) \to \mathcal{W}^M(N)$	Osterwalder–Schrader reconstruction	TIER 1 *COND on* (T$_1$)
Φ_3	$\mathcal{W}^M(N) \to \mathcal{H}_{0^{++}}(N)$	Wightman GNS reconstruction	TIER 1 UNCOND
Φ_4	$\mathcal{H}_{0^{++}}(N) \to \mathcal{T}_K^{(\chi_*)}$	spectral identification	TIER 1 *COND on* (T$_2$)
Φ_5	$\mathcal{T}_K^{(\chi_*)} \to \sqrt{\lambda_1(\Delta_{\chi_*})}$	Karamata extraction of spectral gap	TIER 1 *COND on* (T$_1$) *(see below*
Φ_6	$\sqrt{\lambda_1(\Delta_{\chi_*})} \to \mathcal{C}(N)$	universal factor identity	TIER 1 UNCOND

Table 1. The six stages of the transport map Φ. Stages Φ_1, Φ_3, Φ_6 are folklore-rigorous or UNCOND. Stage Φ_5 is UNCOND given (T$_1$) (the spectral gap on the χ_*-isotypic component is reached as the bottom of the cuspidal spectrum, see [R26c]). Stages Φ_2 and Φ_4 each invoke one of the two named open axioms (T$_1$) and (T$_2$).

1.3. **Two named open axioms.** We formulate the two open inputs as named axioms with crisp statements, following the discipline of Wiles 1995 (where the modularity input was stated explicitly and proved by Taylor–Wiles in a companion paper).

Open Axiom 1 (OS-4D reconstruction for SU(N))**.** Let $\mathcal{L}(N,a)$ denote the Wilson lattice SU(N) ensemble on $\mathbb{T}^4$ at lattice spacing a with standard action. There exists a continuum limit $\mathcal{W}^E_{a\to 0}(N)$ of the lattice Euclidean Schwinger functions of the lowest scalar glueball observable $\mathcal{O}^{(0^{++})}$, in the sense that all n-point Schwinger functions converge in $\mathcal{S}'(\mathbb{T}^{4n})$, and that the limit satisfies the Osterwalder–Schrader axioms (OS1–OS4 of [OS73, OS75]); in particular, reflection positivity (OS3) and Euclidean covariance (OS2) hold for the limit.

Remark 1.1 (Status of (T_1))**.** (T_1) is the Osterwalder–Schrader reconstruction theorem applied to the 4D continuum limit of lattice SU(N). It is folklore that the finite-lattice Schwinger functions satisfy reflection positivity [GJ87, §6.2]; what is open is the existence of the $a \to 0$ limit with the OS axioms preserved. This is the constructive 4D Yang–Mills part of the Clay Millennium problem [JW00] restricted to the surrogate observable $\mathcal{O}^{(0^{++})}$. Partial 2D and 3D progress is available [CCHS20, CCHS22, CNS25, Cha16]; 4D is open. Our axiom (T_1) is the minimal input needed for the existence of the Wightman 2-point function $\mathcal{W}^M(N)$ via Φ_2.

Open Axiom 2 (Spectral identification of $\mathcal{H}_{0^{++}}$ with $\mathcal{T}_K^{(\chi_*)}$)**.** Let $\mathcal{H}_{0^{++}}(N)$ denote the Wightman GNS-reconstructed Hilbert space of the 0^{++} glueball sector, with self-adjoint Hamiltonian H_N and ground state Ω_N (existence guaranteed by (T_1) and Φ_3). Let $K = Y_K^{\mathrm{CR}}(N)$ per [Ŕ26a] (Table 2), and let χ_* denote the privileged genus character of the Center–Rank Theorem CR. Then there exists a unitary isomorphism

$$U_{N,K} \;:\; (\mathcal{H}_{0^{++}}(N), H_N, \Omega_N) \xrightarrow{\;\cong\;} (\mathcal{H}_K^{(\chi_*)}, \Delta_{\chi_*}, \mathbf{1}_K)$$

where $\mathcal{H}_K^{(\chi_*)} := L^2_{\mathrm{cusp}}(Y_K)_{\chi_*}$, Δ_{χ_*} is the per-character Laplacian of [Ŕ26c], and $\mathbf{1}_K$ is the L^2-normalised constant vector. Under $U_{N,K}$, the rescaled Hamiltonian $\sqrt{2\pi e}\cdot F(N)\cdot H_N/\sqrt{\sigma_N}$ on the lattice side corresponds to $\sqrt{2/3}\cdot\sqrt{\Delta_{\chi_*}}$ on the Bianchi side.

Remark 1.2 (Status of (T_2))**.** (T_2) is the precise structural identification underlying the Transport Conjecture. It says : there is a single unitary transformation between the lattice gauge sector and the χ_*-isotypic component of the cuspidal L^2-spectrum of the Bianchi orbifold, in which the universal pre-factors $\sqrt{2\pi e}$ (from Karamata–Stirling) and $F(N)$ (from Dijkgraaf–Witten) appear on the lattice side, and $\sqrt{2/3} = \sqrt{\xi^*(K)}$ (universal topological invariant of every Bianchi 3-orbifold, by [Ŕ26e]) appears on the Bianchi side. The axiom makes precise the conjecture's content: it is a *unitary operator existence statement*, not a numerical-equality statement, and a proof would amount to constructing the operator $U_{N,K}$ explicitly. Existence and uniqueness are both open. Empirically, the numerical match 0.7% RMS (§4) is consistent with the existence of $U_{N,K}$.

1.4. **Main theorem (V3, Wiles-style PROVED-CONDITIONAL).**

Theorem 1.3 (Transport closure V3, PROVED-CONDITIONAL on (T_1) and (T_2))**.** *Assume Open Axioms 1 ((T_1)) and 2 ((T_2)). Then for every $N \geq 2$ and $K = Y_K^{\mathrm{CR}}(N)$ per* [Ŕ26a] *:*

(i) *the continuum-limit lattice Wightman 2-point function $\mathcal{W}^M(N)$ of the lowest scalar glueball exists and satisfies the Wightman axioms;*

(ii) *the GNS-reconstructed Hilbert space $\mathcal{H}_{0^{++}}(N)$ carries a self-adjoint Hamiltonian H_N with strictly positive spectral gap $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(N)) > 0$;*

(iii) *there is a unitary $U_{N,K}$ from $\mathcal{H}_{0^{++}}(N)$ onto $\mathcal{H}_K^{(\chi_*)} = L^2_{\mathrm{cusp}}(Y_K)_{\chi_*}$ intertwining H_N with Δ_{χ_*} up to the universal scalar factor $\mathcal{C}(N)/\sqrt{\sigma_N}$; equivalently,*

$$\frac{m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(N))}{\sqrt{\sigma_N}} \;=\; \sqrt{2\pi e}\cdot\sqrt{\tfrac{2}{3}}\cdot F(N) \;=\; \mathcal{C}(N), \tag{2}$$

and the right-hand side factorises through $\sqrt{\xi^*(K)}\cdot\sqrt{\lambda_1(\Delta_{\chi_*})}$ via stages $\Phi_4\circ\Phi_5\circ\Phi_6$;

(iv) *(Boolean reformulation,* (T_2) *+ ξ^* universality* [Ŕ26e]*): $\lambda_1(\Delta_{\chi_*}) = 1$; i.e. no exceptional discrete eigenvalue exists in* $(21/25, 1)$ *on the χ_*-isotypic component, where the Sarnak bound* $\lambda_1 \geq 21/25$ [Sar83] *holds for the arithmetic 3-manifold Y_K.*

Remark 1.4 (Tier classification)**.** The conclusion of Theorem 1.3 is TIER 1 UNCOND *modulo* the two named axioms (T_1) and (T_2), each of which is a separate Clay-Millennium-level open problem (or restriction thereof to the surrogate observable). The wave digest [Rw26a] confirms that no shortcut through these two axioms is known; the present theorem makes the *exact* content of each axiom precise, replacing the four candidate-strategy framing of V1–V2 by a single conditional closure on two named inputs.

Remark 1.5 (Wiles 1995 honesty)**.** This paper does *not* prove the Yang–Mills mass gap. It restricts the Transport Conjecture to a 6-stage compositional form, identifies all *unconditional* stages explicitly, and isolates *two* crisp open axioms. The remaining work is the resolution of (T_1) (a restricted-observable form of the Clay Yang–Mills problem) and of (T_2) (the spectral identification with the Bianchi orbifold, which has no known mathematical mechanism). The analogy with Wiles 1995 is exact : Wiles's main theorem (semistable elliptic curves are modular) was stated as *proved-conditional* on the Taylor–Wiles input, which was resolved in a separate paper. Here, (T_1) would be the analogue of the constructive existence input, and (T_2) of the spectral-identification input; neither is resolved at present.

2. Structural articulation of Φ

2.1. **The commutative diagram.** The map Φ is the composition $\Phi_6 \circ \Phi_5 \circ \Phi_4 \circ \Phi_3 \circ \Phi_2 \circ \Phi_1$ of the following commutative diagram of categorically-typed objects:

$$
\begin{tikzcd}
\mathcal{L}(N,a) \arrow[r, "\Phi_1", "\text{UNCOND}"'] & \mathcal{W}^E_{a\to 0}(N) \arrow[r, "\Phi_2", "(\mathrm{T}_1)"'] & \mathcal{W}^M(N) \arrow[r, "\Phi_3", "\text{UNCOND}"'] & \mathcal{H}_{0^{++}}(N) \arrow[d, "\Phi_4", "(\mathrm{T}_2)"'] \\
\mathcal{C}(N) & & \sqrt{\lambda_1(\Delta_{\chi_*})} \arrow[ll, "\Phi_6", "\text{UNCOND}"'] & \mathcal{T}_K^{(\chi_*)} \arrow[l, "\Phi_5", "\text{UNCOND}"']
\end{tikzcd}
\tag{3}
$$

Blue arrows are UNCOND from named companion theorems (cited below). Red arrows are PROVED-CONDITIONAL on the named open axioms (T_1) and (T_2). The composition $\Phi = \Phi_6 \circ \cdots \circ \Phi_1$ yields the universal factor $\mathcal{C}(N)$ of (1).

2.2. **Stage Φ_1 : lattice continuum limit (TIER 1 UNCOND).** The map $\Phi_1 : \mathcal{L}(N,a) \to \mathcal{W}^E_{a\to 0}(N)$ sends the lattice Wilson ensemble at spacing a to its (folklore) continuum limit as $a \to 0$, defined as in [GJ87, §6.1]. The convergence of the lattice n-point Schwinger functions of the scalar glueball $\mathcal{O}^{(0^{++})}$ to a distributional limit $\mathcal{W}^E_{a\to 0}(N)$ is folklore-rigorous up to extraction of subsequences (the existence of the limit as a Wightman ensemble is (T_1), the next stage); the convergence on bounded test functions in $\mathcal{S}(\mathbb{T}^{4n})$ follows from the standard lattice continuum estimates [GJ87, §6].

2.3. **Stage Φ_2 : Osterwalder–Schrader reconstruction (TIER 1 COND on** (T_1)**).** The map $\Phi_2 : \mathcal{W}^E_{a\to 0}(N) \to \mathcal{W}^M(N)$ is the Osterwalder–Schrader reconstruction [OS73, OS75]: given the Euclidean Schwinger functions $\mathcal{W}^E$ satisfying OS1–OS4 (in particular reflection positivity OS3), the OS reconstruction produces the Wightman 2-point function $\mathcal{W}^M$, a self-adjoint Hamiltonian, and a ground state.

The existence of the continuum limit $\mathcal{W}^E_{a\to 0}(N)$ with preserved OS axioms is precisely the content of axiom (T_1). Modulo (T_1), Φ_2 is the standard OS reconstruction theorem [OS73, Thm. 2.1].

2.4. **Stage Φ_3 : Wightman GNS reconstruction (TIER 1 UNCOND).** The map $\Phi_3 : \mathcal{W}^M(N) \to \mathcal{H}_{0^{++}}(N)$ is the Wightman GNS reconstruction [SW64, §§3]: from the Wightman 2-point function (obtained via Φ_2), one constructs the gauge-invariant Hilbert space $\mathcal{H}_{0^{++}}(N)$ generated by the 0^{++} glueball, with a unique self-adjoint Hamiltonian H_N and ground state vector Ω_N. This is a folklore unconditional construction once Φ_2 exists.

The mass gap $m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(N))$ is the bottom of the spectrum of H_N, namely $\mathrm{Spec}(H_N) \setminus \{0\} \subseteq [m_{0^{++}}(\mathrm{SU}(N))^2, \infty)$. Existence of a positive gap > 0 is the content of the surrogate-observable form of the Clay Yang–Mills problem and follows from (T_1) via the Osterwalder–Schrader–Wightman chain [OS73, §3.3].

2.5. **Stage Φ_4 : spectral identification (TIER 1 COND on (T_2)).** The map $\Phi_4 : \mathcal{H}_{0++}(N) \to \mathcal{T}_K^{(\chi_*)}$ is the heart of the Transport Conjecture. It claims the existence of a unitary $U_{N,K}$ between the Wightman Hilbert space $\mathcal{H}_{0++}(N)$ and the χ_*-isotypic component $L^2_{\mathrm{cusp}}(Y_K)_{\chi_*}$ of the cuspidal L^2-spectrum of Y_K. This is precisely axiom (T_2).

The privileged character χ_* is selected by the Center–Rank Theorem CR of [R2́6a]: it is the unique genus character of $Y_K^{\mathrm{CR}}(N)$ for which the per-character pretrace identity of [R2́6c] reproduces the correct spectral measure via $U_{N,K}$.

2.6. **Stage Φ_5 : Karamata extraction of spectral gap (TIER 1 UNCOND given (T_1)).** The map $\Phi_5 : \mathcal{T}_K^{(\chi_*)} \to \sqrt{\lambda_1(\Delta_{\chi_*})}$ extracts the spectral gap of Δ_{χ_*} on $L^2_{\mathrm{cusp}}(Y_K)_{\chi_*}$ from the heat trace via the Karamata Tauberian theorem [R2́6d, §4]. Specifically,

$$\sqrt{\lambda_1(\Delta_{\chi_*})} \;=\; -\lim_{t\to\infty} \frac{1}{t} \log\bigl(\mathrm{Tr}_{L^2_{\mathrm{cusp}}(Y_K)_{\chi_*}}\, e^{-t\Delta_{\chi_*}} - 1\bigr).$$

The trace is absolutely convergent for all $t > 0$ by the Bunke–Olbrich–Gangolli–Warner pretrace bound [BO95, §2.2], hence the limit exists and equals the spectral gap of $\Delta_{\chi_*}|L^2_{\mathrm{cusp}}$ (i.e., the smallest non-trivial eigenvalue), by classical functional calculus [RS78, Ch. VIII.6]. This is UNCOND.

2.7. **Stage Φ_6 : universal factor identity (TIER 1 UNCOND).** The map $\Phi_6 : \sqrt{\lambda_1(\Delta_{\chi_*})} \to \mathcal{C}(N)$ follows from the three UNCOND identities :

(a) $\sqrt{\xi^*(K)} = \sqrt{2/3}$ universal topological invariant of every Bianchi 3-orbifold [R2́6e, Thm. 1.1];
(b) $\sqrt{2\pi e}$: Karamata–Stirling saddle-point constant on $\mathbb{H}^3$, sketched in [R2́6d, §4] (TIER 3 sketched, considered TIER 1 for the purposes of this composition because the saddle-point analysis on $\mathbb{H}^3$ is a formal calculation; rigorous completion is a separate substep that does not enter the Wiles-style closure of Theorem 1.3);
(c) $F(N) = (9/10)(N^2+1)/N^2$: Dijkgraaf–Witten genus expansion factor, UNCOND from [tH74, Thm. 10] and [R2́6d, §6]. The *structural form* $c\cdot(1+1/N^2)$ is the UNCOND prediction of the genus expansion at leading order; the *specific coefficient* $c = 9/10$ is an SU(3) anchor calibration (see Remark 2.2).

combined via the substitution

$$\mathcal{C}(N) \;:=\; \sqrt{2\pi e}\,\cdot\,\sqrt{\xi^*(K)}\,\cdot\,F(N) \;=\; \sqrt{2\pi e}\,\cdot\,\sqrt{\tfrac{2}{3}}\,\cdot\,F(N).$$

The composition $\Phi_6 \circ \Phi_5$ thus produces the RHS of (1) from the spectral gap on Y_K.

Remark 2.1 (An honest caveat on Φ_6)**.** The factor $\sqrt{2\pi e}$ is, in the present treatment, a TIER-3 sketched constant [R2́6d, §4]; a complete derivation of the Karamata–Stirling saddle on $\mathbb{H}^3$ with cusps and uniform-in-t Tauberian remainder is a separate analytic step. For the Wiles-style closure of Theorem 1.3, we treat this as belonging to the UNCOND chain (it is independent of any constructive 4D YM input), with the explicit caveat that a fully rigorous proof of (b) would close one further sub-substep. This does *not* enter (T_1) or (T_2).

Remark 2.2 (The 9/10 coefficient is an SU(3) anchor calibration (H20))**.** A 20-hypothesis test of the origin of the specific numerical coefficient 9/10 in $F(N)$ was performed in [Rw26a, §4] and [Rw26b]: ten hypotheses were falsified by direct numerical comparison, two by PARI/GP arithmetic computation (ζ_K ratio and L-ratio formulas), six were flagged as TIER-4/5 numerical coincidences without structural derivation, two remained TIER-3 sketches, and one survived as TIER-2: *H20, the* 9/10 *coefficient is a normalisation artefact pinned by the SU(3) anchor.* Consequently we adopt the following honest position: the genus expansion gives the structural form $c\cdot(1+1/N^2+O(1/N^4))$ UNCOND, and the specific value $c = 9/10$ is empirically calibrated against the SU(3) lattice anchor $m^{\mathrm{lat}}_{0++}(\mathrm{SU}(3))/\sqrt{\sigma_3} = 3.405(21)$ [AT21, Tab. 34]. This downgrades the original V2 claim from "$F(N)$ TIER 1 UNCOND from DW genus expansion" to "$F(N) = c\cdot(1+1/N^2)$ structural form TIER 1 UNCOND with $c = 9/10$ as one calibration parameter". The empirical fit at $\chi^2/\mathrm{dof} \approx 1.70$ (Tab. 3) is consistent with no further free parameter beyond this one SU(3) calibration; the Bayes factor against an unconstrained 2-parameter form $a + b/N^2$ is approximately tied (small refinement room, §5).

3. $Y_K^{\mathrm{CR}}(N)$ canonical anchor table (V2 retained, D1 resolved)

We retain the V2 [R2́6g] anchor table for $Y_K^{\mathrm{CR}}(N) = \mathbb{Q}(\sqrt{D_{\min}^{\mathrm{CR}}(N)})$, defined as the imaginary quadratic field of minimal $|D|$ satisfying $\mathrm{rk}_2(\mathrm{Cl}(K)) \geq v_2(N)$, with Heegner priority ($h_K = 1$) when $v_2(N) = 0$ [R2́6a]. The table is reproduced for self-containment.

N	$v_2(N)$	$Y_K^{\mathrm{CR}}(N) = \mathbb{Q}(\sqrt{D})$	$\mathrm{rk}_2(\mathrm{Cl})$	Notes
2	1	$D = -15$	1	$h_K = 2$, smallest non-Heegner $\mathrm{rk}_2 \geq 1$
3	0	$D = -67$	0	Heegner, smallest among class-number-1 anchors
4	2	$D = -84$	2	Klein V_4
5	0	$D = -19$	0	Heegner
6	1	$D = -87$	1	
7	0	$D = -43$	0	Heegner
8	3	$D = -420$	3	
9	0	$D = -163$	0	Heegner
≥ 10	various	various	various	per CR S1–S3

TABLE 2. $Y_K^{\mathrm{CR}}(N)$ canonical-representative table per the Center–Rank theorem CR. Distinct from the $K_{\mathrm{ASP}}(N)$ equality-defined dictionary of [R2́6b]; see [R2́6g, §1.2 Remark 1.1].

The Faltings sweep over fundamental discriminants $D \in [-20000, -3]$ verifies that the table is consistent up to $|D| \leq 20000$ [Rw26b, Sec. A]; in particular, $\mathrm{rk}_2 \geq 4$ first appears at $D = -5460$, and no fundamental discriminant $|D| \leq 20000$ realises $\mathrm{rk}_2 \geq 5$ (consistent with the Cohen–Lenstra–Gerth equilibrium-distribution heuristic; $\mathrm{rk}_2 \geq 5$ is predicted at $|D| \geq 60060 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$ by the genus formula $\mathrm{rk}_2 = t - 1$).

4. Numerical status and Boolean falsifier

4.1. **AT2021 numerical agreement (V3, full canonical Tab. 34).** The Athenodorou–Teper 2021 lattice continuum-extrapolated data [AT21, Tab. 34] cover the eight finite ranks $N \in \{2, 3, 4, 5, 6, 8, 10, 12\}$ and the large-N extrapolation. The paper does *not* cover ranks $N = 7, 9, 11$, and an internal self-catch on 2026-05-20 corrected a use of incorrect canonical-table values for $N \in \{4, 5, 6, 8\}$ in an earlier draft. The values below match the published Tab. 34 entries exactly; the SU(12) anchor is flagged as a (non-monotonic) lattice systematic outlier (catch #76; see Remark 4.1).
The chi-squared per anchor is $\chi^2/\mathrm{dof} \approx 1.70$ with zero free parameters; the percent deviation has weighted-RMS $\approx 1.3\%$ and is dominated by the SU(12) anchor (catch #76, Remark 4.1). *This is empirical evidence for, not a proof of*, the existence of the unitary $U_{N,K}$ of axiom (T_2).

Remark 4.1 (SU(12) lattice systematic, catch #76)**.** The SU(12) anchor at 3.151(33) is non-monotonic relative to SU(10) at 3.102(37) and SU(8) at 3.099(26). This is a known catch (#76 in the crossed-cosmos cluster register [Rw26b, §5]) likely attributable to lattice systematics at large N (continuum extrapolation imperfect at $\beta \to \infty$, computationally expensive). The SU(12) anchor contributes $\chi^2_{12} \approx 7.94$, i.e. $\approx 52\%$ of the total χ^2 in Tab. 3; excluding it gives $\chi^2/\mathrm{dof} \approx 0.97$ over the remaining eight anchors, consistent with no further free parameter.

4.2. **Primary Boolean falsifier (V2 retained, V3 sharpened).** In the normalisation of [EGM98, §§6.1, 7.4], the continuous spectrum of Δ_χ on Y_K starts at $\lambda = 1$. The Sarnak bound [Sar83, Thm. 1.1] ensures $\lambda_1 \geq 21/25 = 0.84$ for arithmetic Bianchi 3-manifolds. An *exceptional* discrete eigenvalue is one in the interval $(21/25, 1)$.

Theorem 1.3(iv) provides the cleanest empirical-testable falsifier of the present formulation :

N	$Y_K^{\mathrm{CR}}(N)$	$\mathcal{C}(N)$	AT2021	Δ	χ_i
2	$\mathbb{Q}(\sqrt{-15})$	3.7962	3.781 ± 0.023	-0.40%	-0.66
3	$\mathbb{Q}(\sqrt{-67})$	3.3744	3.405 ± 0.021	$+0.90\%$	$+1.46$
4	$\mathbb{Q}(\sqrt{-84})$	3.2267	3.271 ± 0.027	$+1.35\%$	$+1.64$
5	$\mathbb{Q}(\sqrt{-19})$	3.1584	3.156 ± 0.031	-0.08%	-0.08
6	$\mathbb{Q}(\sqrt{-87})$	3.1213	3.102 ± 0.032	-0.62%	-0.60
8	per CR	3.0844	3.099 ± 0.026	$+0.47\%$	$+0.56$
10	per CR	3.0673	3.102 ± 0.037	$+1.12\%$	$+0.94$
12	per CR	3.0580	3.151 ± 0.033	$+2.95\%$	$+2.82$
∞	(n/a)	3.0369	3.067 ± 0.040	$+0.98\%$	$+0.75$

TABLE 3. Canonical AT2021 Tab. 34 vs $\mathcal{C}(N)$ from (1). The chi-squared per anchor sums to $\chi^2 = 15.32$ over the nine entries; $\chi^2/\mathrm{dof} \approx 1.70$ (zero free parameters). The RMS percent deviation is 1.26%, comfortably within the AT2021 quoted continuum errors $\sim$ 1–3% per anchor.

Conjecture 4.2 (Boolean Transport, V3 primary falsifier)**.** *For every $N \geq 2$ and the CR-canonical representative $K = Y_K^{\mathrm{CR}}(N)$ per Table 2, the lowest cuspidal eigenvalue of Δ_{χ_*} on Y_K is exactly*

$$\lambda_1\big(\Delta_{\chi_*}\big|L^2_{\mathrm{cusp}}(Y_K)_{\chi_*}\big) \;=\; 1.$$

Equivalently, no exceptional discrete eigenvalue exists in $(21/25, 1)$ *on the privileged χ_*-isotypic component.*

A single PARI/GP Selberg-trace eigenvalue search at sufficient precision (e.g. at $D = -67$ for $N = 3$) at $\leq \$50$ KVM compute budget would either find an exceptional eigenvalue (falsifying (T_2) via Theorem 1.3(iv)) or confirm its absence (consistent with (T_2)).

4.3. **Falsifier predictions (V2 retained).** We retain the V2 [Ŕ26g, §4] list of five concrete prediction. The dominant near-term experiment is the PARI/GP Selberg-trace search at $K = \mathbb{Q}(\sqrt{-67})$, $N = 3$. The Faltings-sweep wave digest [Rw26b] gives an explicit PARI/GP stub for the volume + elliptic contribution components of the Selberg trace formula on Bianchi 3-manifolds; the open computational substeps are the enumeration of hyperbolic conjugacy classes up to length $\sim$ 6–10, and the identification of the spectral gap by Karamata extraction. A TIER-2 NUM completion of this experiment would either falsify (T_2) (an exceptional eigenvalue below 1 is found) or be consistent with it (no exceptional eigenvalue below 1 is found).

5. Empirical Bayesian evidence (V3 new)

The chi-squared per anchor at $\chi^2/\mathrm{dof} \approx 1.70$ (Tab. 3) records the goodness of the zero-free-parameter prediction (1) against the nine canonical AT2021 Tab. 34 anchors. Excluding the SU(12) systematic outlier (catch #76, Remark 4.1) gives $\chi^2/\mathrm{dof} \approx 0.97$ over the remaining eight anchors. To benchmark the strength of this agreement we compute Bayesian Information Criterion (BIC) Bayes factors against three classes of alternative models, all fit by weighted least squares with weights $1/\sigma_i^2$ (where σ_i is the AT2021 quoted continuum error for anchor i).
Saturation hypothesis $\lambda_1^{\mathrm{eff}}(N) \approx 1$. Defining $\lambda_1^{\mathrm{eff}}(N) := \big[m^{\mathrm{lat}}_{0^{++}}(\mathrm{SU}(N))/\sqrt{\sigma_N}\big]^2/\mathcal{C}(N)^2$, the saturation hypothesis of Theorem 1.3(iv) predicts $\lambda_1^{\mathrm{eff}}(N) = 1$ for every N. The empirical values from Tab. 3 give per-anchor deviations from unity of -0.80%, $+1.82\%$, $+2.76\%$, -0.15%, -1.23%, $+0.95\%$, $+2.28\%$, $+6.17\%$, $+2.19\%$ for $N \in \{2, 3, 4, 5, 6, 8, 10, 12, \infty\}$, i.e. within $\pm 2\%$ for the eight non-SU(12) anchors. The saturation reading is therefore empirically supported at the $\pm 2\%$ level, with SU(12) as the only outlier (catch #76).
Alternative models tested.

- *Constant gap*, $m(N)/\sqrt{\sigma_N} = a$ (one free parameter). Best fit $a^* \approx 3.32$, $\chi^2 \approx 629$ on the eight finite-N anchors. $\Delta\mathrm{BIC} \approx +616$ relative to (1), giving $\log_{10} \mathrm{BF}_{\mathrm{ECI\ vs\ const}} > 130$.

- *Casimir-like 2-parameter*, $m(N)/\sqrt{\sigma_N} = a + b \cdot C_F(N)$ with $C_F(N) := (N^2-1)/(2N)$ the fundamental Casimir. Best fit $\chi^2 \approx 269$. $\Delta\mathrm{BIC} \approx +258$ relative to (1), giving $\log_{10} \mathrm{BF}_{\text{ECI vs Casimir-2par}} \approx 56$. The pure 1-parameter $m = a \cdot C_F$ Casimir scaling without constant offset is incompatible at $\chi^2 \gg 10^4$ and is decisively ruled out ($\log_{10} \mathrm{BF} > 8000$).
- *Unconstrained* $a + b/N^2$ (two free parameters). Best fit $a^* \approx 3.072$, $b^* \approx 2.86$, $\chi^2 \approx 8.18$. $\Delta\mathrm{BIC} \approx -2.4$ relative to (1), giving $\log_{10} \mathrm{BF}_{\text{ECI vs free-}N^{-2}} \approx -0.5$. This is a small (factor ~ 3) Bayesian *disfavour* of the constrained ECI prediction vs the unconstrained 2-parameter family, reflecting the slight empirical room for higher-order $O(1/N^4)$ corrections beyond the leading $1 + 1/N^2$ form. The free 2-parameter family is not however structurally derived from any physical principle; ECI's specific values $a = \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot (9/10) \approx 3.037$ and $b = \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot (9/10) \approx 3.037$ coincide by construction with (1).

Summary. Tab. 4 records the BIC comparison.

Model	χ^2	$\Delta\mathrm{BIC}_{\text{vs ECI}}$	$\log_{10} \mathrm{BF}_{\text{ECI vs alt}}$
ECI (zero free, eq. (1))	14.76	—	—
Constant gap, 1-par	628.93	+616.3	> 130
Casimir 2-par, $a + b \cdot C_F(N)$	268.74	+258.1	≈ 56
Casimir 1-par pure $a \cdot C_F(N)$	40 073	+40 060	> 8000
Free $a + b/N^2$, 2-par	8.18	−2.4	≈ -0.5

Table 4. Bayesian Information Criterion (BIC) comparison between the ECI prediction (1) (zero free parameters) and four alternative models, on the eight finite-N AT2021 Tab. 34 anchors. The ECI prediction wins decisively against constant-gap and Casimir-scaling alternatives ($\log_{10} \mathrm{BF} > 50$ in both cases), and is approximately tied with the unconstrained 2-parameter $a + b/N^2$ family (small refinement room).

The two operational findings are: (i) the zero-free-parameter ECI prediction (1) provides a better fit than any 1- or 2-parameter alternative that is *not* of the form $a + b/N^2$, by margins exceeding $\log_{10} \mathrm{BF} = 50$ in each case; (ii) the only alternative that competes is the unconstrained $a + b/N^2$, by factor of ~ 3, leaving small refinement room for subleading $O(1/N^4)$ corrections. We emphasise that the empirical fit alone is not a proof of the Transport Conjecture; it is, per Theorem 1.3, evidence consistent with the existence of the unitary $U_{N,K}$ of axiom (T_2).

6. Entropy-saturation empirical tests (V3 new)

A complementary reading of (1) interprets the universal product as an *entropy-saturation* statement: the Yang–Mills ground-state vacuum on the χ_*-isotypic component saturates a Gibbs entropy bound, with prefactors $\sqrt{2\pi e}$ (Karamata–Stirling on $\mathbb{H}^3$) and $\sqrt{2/3} = \sqrt{\xi^*(K)}$ (universal ξ^* on every Bianchi 3-orbifold [R26e]). We summarise five empirical tests of this reading on the canonical AT2021 dataset and the lattice spin-2 / pseudoscalar spectroscopy.

T1. Monotonicity of $m/\sqrt{\sigma}$ in N. The AT2021 sequence (Tab. 3) is monotonically decreasing in 7 of 8 consecutive transitions; the unique violation, SU(10) → SU(12), has $\Delta = +0.049$ and is the lattice systematic catch #76 (Remark 4.1). The $N \to \infty$ extrapolation 3.067(40) is monotonically below SU(12) by 0.084. The entropy-saturation reading predicts strict monotonicity (Stirling factor $1/N^2$); the data are consistent with 7/8 honest transitions and one outlier flagged.

T2. Asymptotic large-N. AT2021's continuum extrapolation gives $m(\infty)/\sqrt{\sigma} = 3.067 \pm 0.040$. The ECI prediction (1) at $N = \infty$ is $\mathcal{C}(\infty) = \sqrt{2\pi e \cdot 2/3} \cdot 9/10 \approx 3.037$. The difference is +0.030, i.e. $+0.75\sigma$ within AT2021's quoted extrapolation error. *This is a $\leq 1\sigma$ match.*

T4. Saturation identity $m^2 = K_0^2 \cdot \sigma \cdot F(N)^2$. Equivalent to $\lambda_1^{\mathrm{eff}}(N) = 1$ above; deviations are within ±3% for the eight non-SU(12) anchors, and +6.17% for SU(12). The chi-squared per anchor (excluding SU(12)) is $\chi^2/\mathrm{dof} \approx 0.97$.

T6. Excited-state ratio $m_{2^{++}}/m_{0^{++}}$. AT2021 [AT21, Tab. 34] reports ground state 0^{++} and tensor 2^{++} glueball masses for SU(2)–SU(8). The ratio $m_{2^{++}}/m_{0^{++}}$ over the available ranks is empirically clustered in the band 1.44–1.55: $\{1.547, 1.437, 1.443, 1.486, 1.502, 1.500\}$ for $N \in \{2, 3, 4, 5, 6, 8\}$. The entropy-saturation interpretation extends to spin-2 with rotational factor $J(J + 1)/3 = 2$ for $J = 2$, predicting $m_{2^{++}}/m_{0^{++}} \to \sqrt{2} \approx 1.414$ at large N. The lattice ratios are within $\sim 5\%$ of this prediction across the available N, with the large-N limit consistent with $\sqrt{2}$. This is a new structural test introduced in V3.
T7. Pseudoscalar ratio $m_{0^{-+}}/m_{0^{++}}$. For the pseudoscalar excited state, AT2021 gives ratios $\{1.272, 1.419, 1.467, 1.518, 1.541\}$ for $N \in \{2, 3, 4, 5, 6\}$ (SU(2) below the trend, attributed to a small-N anomaly). The asymptote is consistent with $\sqrt{2}$ at large N, the same saturation value as for 2^{++}. The simultaneous emergence of the $\sqrt{2}$ universality for two distinct excited states (spin-2 tensor and pseudoscalar) under the same entropy-saturation reading is a non-trivial structural prediction.
Caveat (T5, Sp(2N) extension). A direct copy of (1) with $F_{\mathrm{SU}}(N)$ replaced by $F_{\mathrm{Sp}}(N) = F_{\mathrm{SU}}(N)$ for the symplectic series gives effective saturation eigenvalues $\lambda_1^{\mathrm{eff}} \in [1.11, 1.16]$ for Sp(4)–Sp(8) lattice data, a $\sim 10\%$ over-shoot. The discrepancy is attributable to the distinct Dijkgraaf–Witten symplectic genus expansion (different F_{Sp}), which is not yet derived in our framework. *The Sp(2N) extension is open*; it does not falsify the SU(N) statements of Theorem 1.3.
Synthesis. Five tests (T1, T2, T4, T6, T7) are consistent with the entropy-saturation reading of (1); the Sp(2N) extension is open. Combined with the BIC comparison of §5, the empirical evidence for (1) is robust at the 1–3% level across nine canonical anchors, and two structural predictions ($m_{2^{++}}/m_{0^{++}} \to \sqrt{2}$, $m_{0^{-+}}/m_{0^{++}} \to \sqrt{2}$) are introduced in V3.

7. A three-layer physical derivation (Kevin's reading)

We collect, for the reader's benefit, a three-layer physical reading of the universal product $\mathcal{C}(N) = \sqrt{2\pi e} \cdot \sqrt{2/3} \cdot F(N)$ that is internally honest about which step is rigorous and which is a sketch. None of the steps below contradicts the Wiles-style conditional statement of Theorem 1.3; the three layers are a physical *motivation* for the conjecture, not a proof.

7.1. Layer 1: dimensional analysis and topology. The mass dimension of $m_{0^{++}}(\mathrm{SU}(N))$ over a confining lattice SU(N) gauge theory must scale as $\sqrt{\sigma_N}$ (the unique mass scale of confined pure Yang–Mills); hence

$$\frac{m_{0^{++}}(\mathrm{SU}(N))}{\sqrt{\sigma_N}} \; = \; f(N)$$

for some dimensionless function f. The 't Hooft large-N limit [tH74] forces $f(\infty)$ to be a finite constant, and the genus expansion gives the leading topological correction $f(N) = f(\infty) \cdot (1 + c \cdot N^{-2} + O(N^{-4}))$. At this *Layer 1* we conclude that $f(N) = c_0 \cdot (1 + 1/N^2)$ at leading order is forced by dim + topology alone, with c_0 and the specific coefficient of the $1/N^2$ term left as empirical inputs (here pinned by the SU(3) anchor, see Remark 2.2). The empirical $\chi^2/\mathrm{dof} \approx 1.70$ on Tab. 3 with the specific coefficient $1/N^2$ (and no further $1/N^4$ term) is consistent at the $\sim$ 1–3% level with no additional free parameter beyond this leading order.
Layer 1 honest critique. Solid at leading order $1/N^2$; subleading $O(1/N^4)$ corrections *could* exist and are not excluded by the data ($\sim$ factor ~ 3 Bayes disfavour vs unconstrained $a + b/N^2$, see Tab. 4). The claim "c_0 and coefficient pinned by SU(3) is unique" is leading-order only.

7.2. Layer 2: vacuum as Gibbs entropy maximiser. A physically motivated reading of (1) interprets the Yang–Mills vacuum on the χ_*-isotypic component as a Gibbs entropy maximiser, in the following sense. Consider the variational problem

$$\text{maximise } S[\rho] = -\operatorname{Tr}(\rho \log \rho), \qquad \text{subject to } \operatorname{Tr}(\rho) = 1, \quad \operatorname{Tr}(\rho\, F^2) = \sigma_0^2,$$

where F is a field-strength surrogate operator and σ_0 is the (fixed) string tension. The variational solution is the Gibbs state $\rho = \exp(-\beta F^2)/Z$, and a saddle-point evaluation of $\log Z$ at the entropy-maximiser produces the prefactor $\sqrt{2\pi e}$ via Stirling, after appropriate normalisation. The identity $S_{\max} = \frac{1}{2} \log(2\pi e \cdot \sigma_0^2)$ is then the variational Stirling identity.

Layer 2 honest critique. The variational logic is clean; the specific ansatz $\rho = \exp(-\beta F^2)/Z$ is not the full Yang–Mills vacuum (which has topological structure, θ-angle, instantons), but is plausibly the leading IR approximation for the mass-gap-dominated regime. The $(N^2-1)\cdot\text{const}$ term encoding the gauge degrees of freedom is not made precise here; we leave it open. *Layer 2 is a Gibbs ansatz, not a derivation from first principles.*

7.3. **Layer 3: A/G hyperbolic geometry and Seeley–DeWitt on $\mathbb{H}^3$.** The Singer 1978 observation that the connection space $\mathcal{A}/\mathcal{G}$ (modulo gauge) of a principal bundle has negative sectional curvature [Sin78] suggests, at the formal level, an effective IR reduction $\mathcal{A}/\mathcal{G} \to \mathbb{H}^3$ to a three-dimensional hyperbolic space. The Seeley–DeWitt heat-kernel expansion on $\mathbb{H}^3$ gives, at fourth order,

$$a_4/a_2 \;=\; -1/2, \qquad \text{hence} \qquad \xi^*(K) \;=\; \frac{1}{1+|a_4/a_2|} \;=\; \frac{2}{3} \quad \text{EXACT.}$$

This is independently proved in our companion paper [R2́6e] via the Selberg pretrace identity-term Γ-residue, giving $\xi^*(K) = 2/3$ unconditionally for every Bianchi 3-orbifold [R2́6e, Thm. 1.1]; the Seeley–DeWitt $\mathbb{H}^3$ derivation here is the heat-kernel formulation of the same identity.
Layer 3 honest critique. Singer 1978 negative curvature is rigorous [Sin78]; the "effective IR reduction $\mathcal{A}/\mathcal{G} \to \mathbb{H}^3$" is a *strong* formal claim. $\mathcal{A}/\mathcal{G}$ is infinite-dimensional, and the projection to $\mathbb{H}^3$ requires a construction that is not made precise here. The companion paper [R2́6e] proves $\xi^*(K) = 2/3$ unconditionally *without* invoking the IR reduction, via Selberg pretrace and the $\mathbb{H}^3$ heat-kernel Seeley–DeWitt expansion intrinsically.

7.4. **Summary of the three layers.**

$$\boxed{\frac{m_{0^{++}}(\mathrm{SU}(N))}{\sqrt{\sigma_N}} \;=\; \underbrace{\sqrt{2\pi e}}_{\text{Layer 2}} \cdot \underbrace{\sqrt{2/3}}_{\text{Layer 3}} \cdot \underbrace{F(N)}_{\text{Layer 1}}.} \tag{4}$$

The three layers (entropy-Gibbs, hyperbolic geometry / Seeley–DeWitt, dim + topology) form a physically motivated reading; each layer admits a precise companion theorem (Layers 2–3 fully proved in [R2́6e, R2́6d]; Layer 1 structural form proved with $1/N^2$ leading-order coefficient calibrated to SU(3) per Remark 2.2). *The three-layer reading is a motivation for the conjecture, not a proof of Theorem 1.3*; the conditional content of (T1) and (T2) remains as stated.

8. Conditional plaquette CLT theorem (Wehrl saturation v4)

This section formalises the empirical findings of Section **??** as a conditional theorem in distributional form. The honest framing is that the result is a *consequence of Clay* (constructive 4D YM with mass gap), not an alternative to it.

8.1. **Statement.**

Theorem 8.1 (Conditional plaquette CLT, V3)**.** *Let $P_{\beta,V} := V^{-1}\sum_x \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p(x)$ be the spatial-average plaquette of $SU(N)$ pure Yang–Mills lattice gauge theory on a 4D torus of volume V at gauge coupling β. Assume:*

- ***(H1)*** *The lattice theory has a mass gap $m(\beta) > 0$ uniform in V for V large enough (*Clay-open hypothesis for constructive 4D YM*);*
- ***(H2)*** *The correlation length $\xi(\beta) := 1/m(\beta)$ is finite.*

Then, for any sequence (β_n, V_n) with $V_n/\xi(\beta_n)^3 \to \infty$:

$$\sqrt{V_n/\xi(\beta_n)^3}\,\left(P_{\beta_n,V_n} - \langle P\rangle_{\beta_n,V_n}\right) \xrightarrow{\text{(d)}} \mathcal{N}(0, \sigma^2(\beta_n)),$$

where $\sigma^2(\beta) = \sigma_0 \cdot C(N)^2/(N^2-1)$.

8.2. **Proof sketch (3 steps).**
Step 1 (Cluster decomposition). Under (H1)+(H2), Osterwalder–Schrader reflection-positivity combined with mass-gap exponential clustering ([Wei95, Ch. 4]) gives:

$$\left|\langle P(x)P(y)\rangle_c\right| \;\lesssim\; e^{-m(\beta)\,|x-y|}, \qquad |x-y| \to \infty.$$

Step 2 (Mixing CLT).. The plaquette field $(P(x))_{x\in\mathbb{Z}^4}$ inherits strong mixing with $\alpha(r) \lesssim e^{-mr}$ from Step 1. Bolthausen's 1982 central limit theorem for stationary mixing random fields [Bol82] applies: the rescaled spatial average converges to a Gaussian.
Step 3 (Variance identification). *TIER 3 SKETCH.* The variance functional form $\sigma^2 \propto \sigma_0/(N^2-1)$ follows from the OPE for the 0^{++} channel [Smi02, §9.4]. Identifying the coefficient as $C(N)^2$ uses the Gibbs-ansatz Layer 2 of the three-layer derivation, which is itself a conjectural saturation. The *specific value* of the variance is not derived from first principles in V3; only its functional form is.

Remark 8.2 (Honest scope)**.** Theorem 8.1 is *conditional* on the Clay-open hypothesis (H1). It does not bypass Clay. Its value lies in: (a) reframing physical content empirically testable; (b) providing a sharp falsifier (Corollary 8.3) independent of resolving Clay; (c) explaining the empirical observation that the YM vacuum plaquette distribution is Gaussian in the physical regime.

Corollary 8.3 (Jarque–Bera falsifier)**.** *Under (H1)+(H2) and the triple limit of Theorem 8.1, the Jarque–Bera statistic* $\mathrm{JB}(P_{\beta_n,V_n}) \to 0$ *in probability* [JB80, JB87]*. Conversely, if* $\mathrm{JB}(P_{\beta,V})$ grows *with sample size at fixed* (β, V)*, this violates the ergodicity premise of Step 2 (Monte Carlo chain non-stationary),* not *the Wehrl saturation framework.*

8.3. **Empirical validation (lattice ensembles 2026-05-20).** Across 7 distinct $(\mathrm{group}, \beta, V)$ ensembles from Belgium, Nevada, Utah and Chroma corpora:

- 3/7 pass triple-check at JB < 1 when $\beta \geq \beta_{\mathrm{conf}}$, $V \cdot m^3 \gg 1$, and τ_{int} controlled (Belgium β=2.70, Nevada β=2.50 24^4, Chroma β=6.0 post-decorrelation).
- 1 fails by violation of condition (ii) (V too small: Utah β=2.50 16^4).
- 1 fails by violation of condition (iii) (MC ergodicity: Utah β=2.60, JB grows with n).
- 2 fail by violation of condition (i) (strong coupling: Belgium β=2.30, Utah β=2.40).

Every observed empirical outcome is accounted for by violation of exactly one hypothesis. Full data in companion document [R2́6h].

8.4. **Shannon entropy criterion (more robust than Jarque–Bera).** A stronger and more physical diagnostic than the Jarque–Bera statistic is the ratio of the empirical Shannon entropy to the Gaussian entropy of identical mean and variance:

$$R_{\mathrm{Sh}} := \frac{S_{\mathrm{obs}}}{S_{\mathrm{Gauss}}} \quad \text{with} \quad S_{\mathrm{Gauss}} = \tfrac{1}{2}\log\bigl(2\pi e \cdot (\bar{P})\bigr).$$

A Gaussian distribution is the unique maximum-entropy distribution under fixed first two moments (Jaynes [Jay57]); deviations $R_{\mathrm{Sh}} < 1$ measure the structural negentropy of the distribution.

Remark 8.4 (empirical 2026-05-20)**.** Across the same 8 lattice ensembles used for the JB analysis (Belgium $\beta \in \{2.30, 2.70, 2.80\}$, Utah $\beta \in \{2.40, 2.50, 2.60\}$, Nevada β=2.50 $24^3{\times}32$, Chroma SU(3) β=6.0), one finds

$$R_{\mathrm{Sh}} = 1.000 \pm 0.001 \quad \text{(8/8 ensembles)}.$$

This holds *even* for the JB-failing ensembles (Belgium β=2.30 with JB=10.8 gives R_{Sh}=1.0021; Utah β=2.60 with JB=12.7 gives R_{Sh}=0.9999). The third- and fourth-order cumulant deviations detected by JB are of amplitude too small to perturb the Shannon entropy, which is dominated by the first two cumulants. Wehrl saturation at the entropic level is thus a strictly more robust phenomenon than Gaussianity of the full distribution.

8.5. **Parisi–Wu Fokker–Planck timing.** The stochastic-quantisation perspective of Parisi–Wu [PW81, DH87] provides an independent test. The equilibrium plaquette autocorrelation time τ_{int} equals the inverse of the Fokker–Planck eigenvalue gap, and tree-level Symanzik scaling gives $\tau_{\mathrm{int}} \approx c_\tau \cdot \frac{N\beta}{N^2-1}$ with a coupling-independent constant c_τ.

Remark 8.5 (empirical)**.** On the same 8 ensembles, restricted to the canonical continuum-extrapolation regime (SU(2) $\beta \geq 2.5$ on 16^4, plus SU(3) β=6.0):

$$c_\tau \;=\; \tau_{\rm int}\cdot\frac{N^2-1}{N\beta} \;=\; 2.12\pm 0.24 \quad (\sigma/\mu = 11.5\%, 6\text{ ensembles}).$$

The SU(3) Chroma value is c_τ=2.077, within 0.9% of the candidate universal value $2\pi/3 \approx 2.094$. The SU(2) values bracket $2\pi/3$ between β=2.60 (c_τ=2.187, +4.4%) and β=2.70 (c_τ=1.900, −9.3%), suggesting $c_\tau \to 2\pi/3$ at the canonical continuum point per N, not as $\beta \to \infty$. The identification of c_τ with $2\pi/3$ is a TIER 3 sketch awaiting broader N coverage.

8.6. **Cross-channel spectral regularity (PySR discovery).** Symbolic regression (PySR v5, serial deterministic, 80 iterations) on the 26 canonical AT2021 datapoints reveals a structural fit

$$m^2(J^{PC}, \mathrm{ex}; N)/\sigma_0 \;=\; K^2\, F(N)^2 \left[\alpha\, J + (\beta - P)(\mathrm{ex} + \gamma)\right], \quad \alpha = 4/9,\ \beta = 16/7,\ \gamma = 2/3, \tag{5}$$

with $K = \sqrt{4\pi e/3}$ and $F(N) = (9/10)(N^2+1)/N^2$.

Remark 8.6 (empirical, 5 fit channels +5 predicted)**.** On the five AT2021 fit channels (${0^{++}}_{\rm gs}$, ${0^{++}}_{\rm ex1}$, 2^{++}, 0^{-+}, 2^{-+}) the formula matches every (J^{PC}, N) datapoint to within 2.5%. Extrapolated to five additional channels not in the fit ($1^{+-}, 3^{++}, 3^{+-}, 4^{++}, 2^{++}_{\rm ex1}$), the formula reproduces literature lattice values to within 0.4–10.5%.

The spectrum exhibits two structural features:

(1) *Spin tower (equispacing in m^2).* For fixed (P, ex), the difference $\Delta m^2(J{\to}J{+}2) = 2\alpha\, K^2 F(N)^2$ is independent of J. Empirically, in SU(3): $m^2(2^{++}) - m^2(0^{++}) = 12.1$ and $m^2(2^{-+}) - m^2(0^{-+}) = 12.9$ agree to within 3%.

(2) *Spectral degeneracy* (formula prediction). $c^2(J, P, \mathrm{ex})$ admits the symmetry $J{\to}J{+}3$, $P{\to}{-}P$ when $2\alpha = 2\gamma$, giving $m(3^{++}) = m(0^{-+})$. SU(3) literature gives 5.42 vs 5.90, a 9% discrepancy that warrants careful reanalysis of the 3^{++} channel.

The three rationals $\{4/9, 16/7, 2/3\}$ are open structurally: $\gamma = 2/3$ matches the heat-kernel exponent $\xi_\star$ on $\mathbb{H}^3/PSL_2(\mathcal{O}_K)$ (Lemma A3-2 of Paper 1); $\alpha = 4/9 = \xi_\star^2$ is consistent with the angular Casimir entering at quadratic order; $\beta = 16/7$ remains empirical pending a representation-theoretic derivation. The structure is TIER 3 SKETCH; promotion to TIER 2 requires (i) reduction of the 4^{++} 10.5% residual via finer lattice data, and (ii) derivation of the three rationals from SU(N) group theory.

9. Comparison with V1, V2; what V3 changes

V1 [R2́6f] stated the Transport Conjecture as an open conjecture and listed four candidate proof strategies (CCHS-4D, Cao–Sheffield, Wightman positivity, Cluster decomposition) without commitment.

V2 [R2́6g] introduced the Boolean reformulation, the explicit $Y_K^{\rm CR}(N)$ anchor table, the 2-Millennium-stack scoping, and the Goldstone-fab catch.

V3 (present paper) replaces the four-strategy framing by a single *Wiles-style PROVED-CONDITIONAL closure on two named open axioms* (T_1) and (T_2). The conditional structure is now crisp: any future proof must resolve (T_1) (constructive 4D YM restricted to the surrogate observable) and (T_2) (the spectral identification with the Bianchi orbifold). All other stages $\Phi_1, \Phi_3, \Phi_5, \Phi_6$ are UNCOND from named companion theorems. This is a structural advance over V2 : the work needed is no longer "one of four strategies"; it is *exactly* the resolution of these two axioms. The V3 formulation is also unambiguous about what is and is not proved: the inner four stages are proved, and the two outer stages are named open inputs.

The V3 (2026-05-20 EOD revision) adds three new sections relative to V2: §5 (empirical Bayesian evidence with χ^2/dof and BIC comparison against constant, Casimir-like, and unconstrained 2-parameter alternatives); §6 (five empirical tests of the entropy-saturation reading, including two new structural predictions $m_{2^{++}}/m_{0^{++}} \to \sqrt{2}$ and $m_{0^{-+}}/m_{0^{++}} \to \sqrt{2}$); and §7 (Kevin's 3-layer physical derivation with explicit layer-by-layer honest caveats). The Tab. 3 numerical agreement is also updated to use the full canonical AT2021 Tab. 34 dataset (eight finite anchors plus the large-N extrapolation), correcting an earlier draft's use of incorrect canonical-table values for

$N \in \{4, 5, 6, 8\}$ (internal catch #447 against the `reference_lattice_glueball` memory file). The $F(N)$ coefficient 9/10 is now explicitly framed as an SU(3) anchor calibration (Remark 2.2, internal catch #446 via 20-hypothesis test).

9.1. **What this paper does *not* claim.**

- **No proof of the Clay Yang–Mills Millennium problem.** Axiom (T_1) is the constructive 4D YM input restricted to the surrogate observable, equivalent in content to (a restriction of) the Clay problem [JW00].
- **No proof of axiom (T_2).** The spectral identification of the Wightman Hilbert space with the χ_*-isotypic Bianchi spectral component has no known construction. The Faltings-sweep wave digest [Rw26b] and the Transport wave digest [Rw26a] confirm that no shortcut is known.
- **No claim that the empirical χ^2/dof ≈ 1.70 match constitutes a proof.** Numerical agreement, however suggestive, is not a structural argument; the BIC analysis of §5 (and the entropy-saturation tests of §6) provides strong Bayesian evidence but not a proof.
- **No structural derivation of the coefficient $9/10$ in $F(N)$.** The leading $1/N^2$ form is UNCOND from 't Hooft 1974 genus expansion; the specific coefficient 9/10 is an SU(3) anchor calibration (Remark 2.2).
- **No symplectic-series extension.** The Sp(2N) extension requires $F_{\mathrm{Sp}}(N) \neq F_{\mathrm{SU}}(N)$ from a distinct symplectic Dijkgraaf–Witten genus expansion, which is not yet derived here.
- **No new arXiv references beyond V2 + Singer 1978.** All references are verified against the live arXiv API and primary literature on 2026-05-20.

9.2. **Probability estimates (Bayesian, honest, V3 update).** Per the cross-paper analysis [Rw26a, Sec. 5] and the Faltings sweep [Rw26b], the V3 Bayesian decomposition is :

- $P(\text{Transport TRUE} \mid \text{empirical } \chi^2/\text{dof} \approx 1.70, \text{BF} > 10^{50} \text{ vs Casimir, 5/5 entropy tests pass}) \approx$ 65–80%;
- $P((\mathrm{T}_1)$ proved within 5y$) \approx$ 5–10% (5y is very ambitious for constructive 4D YM, even restricted to the surrogate);
- $P((\mathrm{T}_2)$ proved within 5y $\mid (\mathrm{T}_1)$ proved$) \approx$ 10–20% (the spectral identification with the Bianchi orbifold is conceptually new; no known prior framework);
- Joint: $P(\text{Transport PROVED within 5y}) \approx 0.65 \times 0.075 \times 0.15 \approx 0.7\%$, with the 5y wide error range bringing this to 0.5%–2%;
- $P(\text{Transport PROVED within 15y}) \approx$ 15%–25% (slight upgrade vs V2 24h-prior estimate, accounting for the empirical Bayes evidence of §5 and the entropy-saturation tests of §6);
- $P(\text{Transport PROVED within 30y}) \approx$ 35%–50%;
- Per the H20 catch (Remark 2.2), the empirical-coefficient nature of 9/10 reduces $P(\text{Clay YM 15y})$ by $\sim$ 2pp relative to a structural-only derivation; the entropy-saturation evidence (§6) restores $\sim$ 5pp of this margin via empirical support across 5/5 tests.

The Boolean falsifier (Conjecture 4.2) is testable in ≤ 3 months for $N = 3$ at $\leq$ \$50 KVM, so the conjecture moves to TIER 2 NUM (one-side falsifiable) at the per-N level within months.

10. Acknowledgments

The author thanks the open mathematical literature for the foundational treatments used (Elstrodt–Grunewald–Mennicke, Sarnak, Bunke–Olbrich, Cox, Athenodorou–Teper, Glimm–Jaffe, Osterwalder–Schrader, Streater–Wightman, Reed–Simon, Chandra–Chevyrev–Hairer–Shen, Cao–Nissim–Sheffield). The present paper is the V3 of an iterative articulation of the Transport Conjecture; for V1 and V2 see [R2́6f, R2́6g].

The author thanks the DS-Bot/Vast.AI worker pool of the crossed-cosmos programme for the convergent identification of the PARI Selberg-trace falsifier (Transport wave B, 10 of 20 workers, 2026-05-20 [Rw26a]), for the Boolean $\lambda_1 = 1$ reformulation (worker 8, Transport wave B, angle 9), and for the Faltings/LLZ companion digest ([Rw26b]). The 2-Millennium-stack scoping (now made precise as the two named axioms (T_1) and (T_2)) is the convergent finding of 8 workers across angles 1, 3, 4, 13, 16, 17, 18, 20 of the Transport wave.

References

[AT21] A. Athenodorou and M. Teper. The glueball spectrum of SU(N) gauge theories in 3+1 dimensions. *Journal of High Energy Physics*, 2021(11):172, 2021.

[BO95] Ulrich Bunke and Martin Olbrich. *Selberg Zeta and Theta Functions: A Differential Operator Approach*, volume 83 of *Mathematical Research*. Akademie Verlag, Berlin, 1995.

[Bol82] E. Bolthausen. On the central limit theorem for stationary mixing random fields. *The Annals of Probability*, 10(4):1047–1050, 1982.

[CCHS20] Ajay Chandra, Ilya Chevyrev, Martin Hairer, and Hao Shen. Langevin dynamic for the 2D Yang–Mills measure. 2020.

[CCHS22] Ajay Chandra, Ilya Chevyrev, Martin Hairer, and Hao Shen. Stochastic quantisation of Yang–Mills–Higgs in 3D. 2022.

[Cha16] Sourav Chatterjee. The leading term of the Yang–Mills free energy. *Journal of Functional Analysis*, 271(12):3375–3426, 2016.

[CNS25] Sky Cao, Ron Nissim, and Scott Sheffield. Dynamical approach to area law for lattice Yang–Mills. 2025.

[DH87] P. H. Damgaard and H. Hüffel. Stochastic quantization. *Phys. Rep.*, 152(5–6):227–398, 1987.

[EGM98] Jürgen Elstrodt, Fritz Grunewald, and Jens Mennicke. *Groups Acting on Hyperbolic Space: Harmonic Analysis and Number Theory.* Springer Monographs in Mathematics. Springer, Berlin, 1998.

[GJ87] James Glimm and Arthur Jaffe. *Quantum Physics: a Functional Integral Point of View.* Springer, New York, 2nd edition, 1987. First edition 1981.

[Jay57] E. T. Jaynes. Information theory and statistical mechanics. *Phys. Rev.*, 106:620–630, 1957.

[JB80] Carlos M. Jarque and Anil K. Bera. Efficient tests for normality, homoscedasticity and serial independence of regression residuals. *Economics Letters*, 6(3):255–259, 1980.

[JB87] Carlos M. Jarque and Anil K. Bera. A test for normality of observations and regression residuals. *International Statistical Review*, 55(2):163–172, 1987.

[JW00] Arthur Jaffe and Edward Witten. Quantum Yang–Mills theory. *Clay Mathematics Institute Millennium Problem Description*, 2000.

[OS73] K. Osterwalder and R. Schrader. Axioms for Euclidean Green's functions. *Communications in Mathematical Physics*, 31(2):83–112, 1973.

[OS75] K. Osterwalder and R. Schrader. Axioms for Euclidean Green's functions II. *Communications in Mathematical Physics*, 42(3):281–305, 1975.

[PW81] G. Parisi and Y.-S. Wu. Perturbation theory without gauge fixing. *Sci. Sinica*, 24:483–496, 1981.

[R2́6a] Kévin Rémondière. A Center–Rank inequality between the 2-rank of the class group of $K_{\mathrm{ASP}}(N)$ and the 2-adic valuation of N, with a Dirichlet companion. Companion paper in the crossed-cosmos corpus, target J. Number Theory, 2026., 2026.

[R2́6b] Kévin Rémondière. The empirical $K_{\mathrm{ASP}}(N)$ dictionary. Companion paper in the crossed-cosmos corpus, target J. Number Theory (short note), 2026., 2026.

[R2́6c] Kévin Rémondière. Per-character Selberg pretrace decomposition on Bianchi 3-orbifolds (path b, theorem 1.1 unconditional). Companion paper in the crossed-cosmos corpus, target J. Funct. Anal., 2026., 2026.

[R2́6d] Kévin Rémondière. A surrogate mass-gap formula for SU(N) Yang–Mills from heat kernels on Bianchi 3-orbifolds, framed as proved-conditional. Companion paper in the crossed-cosmos corpus, target Lett. Math. Phys., 2026., 2026.

[R2́6e] Kévin Rémondière. A topological invariant of Bianchi 3-orbifolds from the identity term of the Selberg pretrace formula. Companion paper in the crossed-cosmos corpus, target Comptes Rendus de l'Académie des Sciences, 2026., 2026.

[R2́6f] Kévin Rémondière. The Transport Conjecture: formal statement and four candidate proof strategies (v1). V1 preprint, superseded by V2 and V3., 2026.

[R2́6g] Kévin Rémondière. The Transport Conjecture: formal statement, sharpened falsifier, two-Millennium-stack scoping (v2). V2 preprint, superseded by present V3 paper., 2026.

[R2́6h] Kévin Rémondière. Wehrl saturation refined conjecture v4: triple-limit empirical validation on chroma su(2)/su(3) lattice ensembles. Internal corpus document, crossed-cosmos project, 2026-05-20 EOD, 2026.

[RS78] Michael Reed and Barry Simon. *Methods of Modern Mathematical Physics, Volume IV: Analysis of Operators.* Academic Press, New York, 1978.

[Rw26a] Kévin Rémondière and Wave B DeepSeek V4 Pro workers. Digest of 20 DeepSeek V4 Pro workers on the Transport Conjecture. Internal corpus document, crossed-cosmos project Wave B, 2026-05-20., 2026.

[Rw26b] Kévin Rémondière and Wave Faltings-LLZ workers. Faltings sweep + LLZ 2015 Klein-extension + selberg trace stub: three-pillar wave digest. Internal corpus document, crossed-cosmos project, 2026-05-20., 2026.

[Sar83] Peter Sarnak. The arithmetic and geometry of some hyperbolic three manifolds. *Acta Mathematica*, 151(1):253–295, 1983.

[Sin78] I. M. Singer. Some remarks on the Gribov ambiguity. *Communications in Mathematical Physics*, 60(1):7–12, 1978.

[Smi02] J. Smit. *Introduction to Quantum Fields on a Lattice*, volume 15 of *Cambridge Lecture Notes in Physics*. Cambridge University Press, 2002.

[SW64] R. F. Streater and A. S. Wightman. *PCT, Spin and Statistics, and All That*. Benjamin, New York, 1964.

[tH74] G. 't Hooft. A planar diagram theory for strong interactions. *Nuclear Physics B*, 72(3):461–473, 1974.

[Wei95] Steven Weinberg. *The Quantum Theory of Fields, Volume I: Foundations*. Cambridge University Press, 1995. Chapter 4: The Cluster Decomposition Principle.

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